

Functional Requirements

Functional Requirements for Campus Users (Complainants)

1. Authentication & Role-based Access

- The system shall authenticate students, faculty and staff using institute credentials before any complaint activity is allowed.
- The system shall give access based on the user role and shall check the role on the server.

2. Complaint Submission

- Complainants shall be able to submit a complaint with a title, description, category, location (block/room/facility) and date of occurrence.
- Complainants shall be able to attach a photo or supporting document as evidence, with configured file type and size limits.

3. Complaint Tracking ID

- The system shall generate a unique tracking ID for every accepted complaint and show it immediately after submission.

4. Complaint Tracking & History

- Complainants shall be able to view the current status and complete status history of their own complaints.
- Complainants shall be able to search their complaints using tracking ID, category or date.

5. Comments & Additional Information

- Complainants shall be able to exchange comments with the department authority and reply to requests for more information.

6. Reopening & Feedback

- Complainants shall be able to reopen a complaint within the allowed time if the resolution is not satisfactory.
- Complainants shall be able to give feedback when closing a complaint.

Functional Requirements for Department Authorities

1. Complaint Handling

- Department authorities shall be able to view registered complaints, update their status as Registered, In Progress, Resolved or Rejected, and record a resolution or rejection remark.

2. Complaint Search & Filtering

- Authorised department users shall be able to search and filter complaints by category, status, priority, location, department and date range.

3. Reassignment

- A department authority with reassignment rights shall be able to reassign a complaint to another department and record the reason.

4. Complaint Routing

- The system shall route each complaint to the department that owns its confirmed category using administrator-configured routing rules.

5. Complaint Analytics

- Authorised department users shall have access to complaint analytics by category, department, status and location, including average resolution time and recurring complaints.

Functional Requirements for System / Complaint Administrator

1. Role-based Access Control

- The system shall enforce role-based access for Complainant, Department Authority and Administrator, with all role checks done on the server.

2. System Configuration

- Administrators shall be able to configure categories, departments, routing rules, roles and escalation thresholds without changing application code.

3. Duplicate Complaint Detection

- The system shall flag a newly submitted complaint as a possible duplicate when a similar open complaint exists for the same category and location.

4. Authorised Search & Filtering

- Administrators shall be able to search and filter complaints by category, status, priority, location, department and date range.

5. Confidentiality Configuration

- Administrators shall be able to set which complaint categories need confidential handling. For these categories, the system shall hide the complainant identity from the department as required by institute policy.

Non-Functional Requirements

1. Usability

- A student or staff member shall be able to submit a complaint without training or a manual. This can be checked by a short test with first-time users.

2. Performance

- Common pages such as the submission form, My Complaints and the department queue shall load within about 2 seconds on the campus network under normal load.

3. Security

- Passwords shall never be stored in plain text. Authentication and role checks shall be done on the server, and access shall use HTTPS.

4. Privacy

- A complainant shall see only their own complaints. A department authority shall see only complaints assigned to that department. An administrator shall be limited to the access allowed by the role.

5. Reliability

- A submitted complaint shall not be lost due to a temporary network or interface failure. The user shall receive a confirmation with the tracking ID.

6. Scalability

- The system shall support growth in users, complaints and attachments, with an indicative target of 5,000 registered users and 20,000 complaints per year without redesign.

7. Compatibility

- The interface shall work on current mobile and desktop browsers without horizontal scrolling.

8. Auditability

- Every important action shall record the acting user, the action, the affected complaint and a timestamp. Audit entries shall not be editable.

9. Explainability

- Where practical, an ML prediction shall show the predicted class and a confidence indicator so that a human reviewer can decide whether to override it.

10. Data Integrity

- A complaint's status changes shall follow the defined lifecycle only. A failed operation shall not leave a complaint partly updated.

11. Availability

- The service shall be available during campus working hours, with maintenance scheduled outside those hours. An indicative target is 99% monthly availability.

12. Accessibility

- Text, contrast, labels and keyboard navigation shall follow common accessibility practices so that the form is usable by all campus users.

Domain Requirements

1. Complaint Categories and Ownership

- Complaint categories shall match the actual departments of the institute, and each category shall have an assigned department. Electrical and mechanical complaints shall remain separate categories.

2. Complaint Closure

- A complaint shall not be closed without a resolution remark. When the department finishes the work, the complaint shall become Resolved. Final closure shall be done by the complainant or an authorised administrator.